Who Earns Credit in a Forecasting Pool? Source-Level Shapley Values, Core Stability, and Deployment-Faithful Validation

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Abstract

When small firms pool their sales histories to train a shared demand forecaster, someone has to decide who earned the improvement. The Shapley value is the standard answer, but a firm's Shapley value is only as good as the measurement of how much each coalition of firms gains, and that measurement depends on how forecasts are validated. I treat ten Walmart stores from the M5 data as ten firms of different sizes, compute the out-of-sample gain of every one of the 1,023 possible coalitions, and value each store exactly under two protocols: random 5-fold validation, common in data valuation, and walk-forward validation, which mirrors deployment. Both are scored on the same 117 weeks. Four findings follow. (1) Random folds understate what pooling is worth: with a LightGBM forecaster the pool's total value halves (0.147 against 0.310), and the average store's gain falls from 6.3% to 3.1%. The understatement appears in all five games I ran, because leaked future rows help a store's standalone model about 3.5 times more than they help the pooled model. (2) The protocol changes who counts as a contributor: one store moves from roughly zero credit to the second-largest share, and some store's credit changes sign in four of the five games. (3) Shapley allocations are never stable. They fall outside the core in all ten game-protocol pairs, and the core is empty in nine; the one non-empty core appears only under walk-forward validation. (4) Credit drifts: rankings computed on one quarter barely predict the next. Monte Carlo Shapley with 200 permutations recovers the exact ranking (Kendall $\tau \geq 0.95$), while leave-one-out values miss magnitudes by 25% to 66%. For a real forecasting commons, these results argue for walk-forward valuation, reported uncertainty, and an allocation rule that is checked for stability rather than assumed fair.

1 Motivation

I came to this question from a practical problem. In Critical Mass, a buying cooperative our team designed for small Pakistani importers, member firms would pool order histories so that a shared model can forecast demand none of them could forecast well alone. The next question from every firm was the same: if the pooled forecast saves money, how much of that saving is mine? Shapley values give a principled answer: average each firm's marginal contribution over every order in which the pool could have formed. They are efficient, symmetric and widely used in data valuation (Ghorbani & Zou, 2019; Jia et al., 2019).

The answer, however, rests on a number that is easy to get wrong: how much a coalition of firms actually gains from pooling. That gain has to be estimated out of sample, and the estimate depends on the validation protocol. In earlier work on air-quality forecasting in Lahore, I found that switching from random to walk-forward validation reversed which model looked best. Here I ask whether the same switch changes who looks valuable in a data pool. Four questions organise the study:

1. **Q1.** Does pooling help, and whom?
2. **Q2.** Do Shapley values and their ranking change between random and walk-forward validation?
3. **Q3.** Is there a stable allocation (a non-empty core), and does Shapley belong to it?
4. **Q4.** Do values drift over time, and can cheap estimators replace exact computation?

2 Related work

Data Shapley (Ghorbani & Zou, 2019; Jia et al., 2019) and Data Banzhaf (Wang & Jia, 2023) value individual training points; incentive-aware collaborative learning (Sim et al., 2020) and federated contribution measurement (Wang et al., 2020) value data owners; attribution has also been proposed as the basis for paying data creators (Deng et al., 2023). Closest in spirit, regression markets (Pinson et al., 2022) use Shapley allocations to pay for other agents' data in wind-power forecasting and separate in-sample from chronological out-of-sample payments, but they value features supplied to one central forecaster rather than firms pooling series into a shared model, and they do not compare validation protocols or study stability. Two recent papers sharpen the motivation: Shen et al. (2026) show that the structure of the validation set alone can shift Shapley values, but only for in-distribution, sample-level settings without time; Pang et al. (2026) introduce time-aware sample valuation for time series, on classification tasks and without a protocol comparison. Diehl et al. (2025) document how much Shapley-based data valuation depends on arbitrary choices. On the forecasting side, global models trained across many series often beat local ones (Montero-Manso & Hyndman, 2021), which is what makes pooling attractive; and k-fold cross-validation can be valid for evaluating purely autoregressive models (Bergmeir et al., 2018). That last result concerns the error of a single model; this study asks what happens when models trained on different amounts of data are compared, which is exactly what a coalition game does.

3 Setup

Firms and data. The M5 competition data (Makridakis et al., 2022) contain daily unit sales for 3,049 products in ten Walmart stores across California (4), Texas (3) and Wisconsin (3). Each store is one "firm". To mimic a pool of firms of different sizes, each store contributes a random sample of 12, 20, 30, 45 or 60 products (two stores per size, assignment shuffled), drawn from products selling at least one unit a week on average. Sales are aggregated to weeks (277 weeks). The first 104 weeks are a burn-in used only to scale each series and to compute the naive one-step error used for scaling. I repeat the item-level design with three random samples, and add a coarser design in which each store contributes its seven department totals.

Forecaster. One-step-ahead weekly forecasts from a pooled model trained on all rows of a coalition's members. Features are ten lags of the scaled series (1 to 52 weeks), 4- and 13-week means, yearly seasonality, the state's SNAP days and event days in the target week, and the department. The main model is a ridge-regularised linear autoregression, the standard global model in the forecasting literature; because its fit depends only on sums of per-store Gram matrices, all 1,023 coalitions can be refitted exactly in seconds. A LightGBM model checks whether conclusions hold for a nonlinear learner; to keep it tractable it runs on six stores (two per state, 63 coalitions).

The game. For a coalition S, the value is the total error reduction its members get from pooling: $v(S) = \Sigma_{i \in S}\ [\mathrm{err}_i(\text{alone}) - \mathrm{err}_i(S)]$, where $\mathrm{err}_i(S)$ is the mean squared scaled error (MSSE) on store i's own products of the model trained on S's data. Single stores have value zero. Shapley values are computed exactly from all coalitions; the core and least core come from a linear programme over the 1,022 proper coalitions.

Two protocols, one evaluation window. Under **random 5-fold validation (R)**, rows are assigned to folds at random and each fold is predicted by a model trained on the other four. Under **walk-forward validation (W)**, nine quarterly origins each train on all earlier weeks and forecast the next 13. Both protocols are scored on exactly the same 117 weeks, so differences come from the protocol and not from the test period. Uncertainty comes from a moving-block bootstrap over weeks (4-week blocks, 1,000 resamples), applied to the same resampled weeks in both protocols.

4 Results

Game	Pool value $v(N)$		Mean store gain		Stores with $\varphi < 0$		Core non-empty		Shapley in core	τ, R vs W [95% CI]	Sign flips
	R	W	R	W	R	W	R	W			
Departments, linear	−0.442	−0.012	−1.1%	0.4%	5	6	no	no	no	0.78 [0.24, 0.91]	CA_3
Items 1, linear	0.008	0.078	0.4%	1.4%	4	3	no	no	no	1.00 [0.69, 1.00]	CA_2
Items 2, linear	−0.017	0.007	−0.6%	−0.4%	4	4	no	no	no	0.60 [0.47, 0.91]	none
Items 3, linear	0.102	0.111	0.8%	0.9%	2	1	no	no	no	0.73 [0.33, 0.87]	CA_2
Items 1, LightGBM (6 stores)	0.147	0.310	3.1%	6.3%	2	0	no	**yes**	no	0.73 [0.07, 0.87]	CA_2, WI_2

Table 1. Summary of the five pooling games ("Items 1" to "Items 3" are the three random product samples). *v(N)* is the grand coalition's total MSSE reduction; mean store gain is the average percentage error reduction from joining the full pool. "Sign flips" lists stores whose Shapley value φ changes sign between random (R) and walk-forward (W) validation.

4.1 Pooling helps only when the model can use it (Q1)

With the linear global model, pooling is worth very little. The average store gains between −0.4% and 1.4%, at most two of ten stores gain significantly under walk-forward validation in each item sample, and between 96 and 649 of the 1,023 coalitions have negative value, meaning their members would have forecast better alone. At the department level, where each store's data are smooth aggregates, pooling hurts under both protocols, and sharply under random validation. The nonlinear model is different: under walk-forward validation, LightGBM cuts error by 4.7% to 10.7% for five of the six stores, each significant, with WI_2 the one store that gains nothing. Whether a pool is worth forming therefore depends as much on the learner as on the data; a linear model that each store can already fit on its own has little to gain from its neighbours.

4.2 Random folds understate what the pool is worth

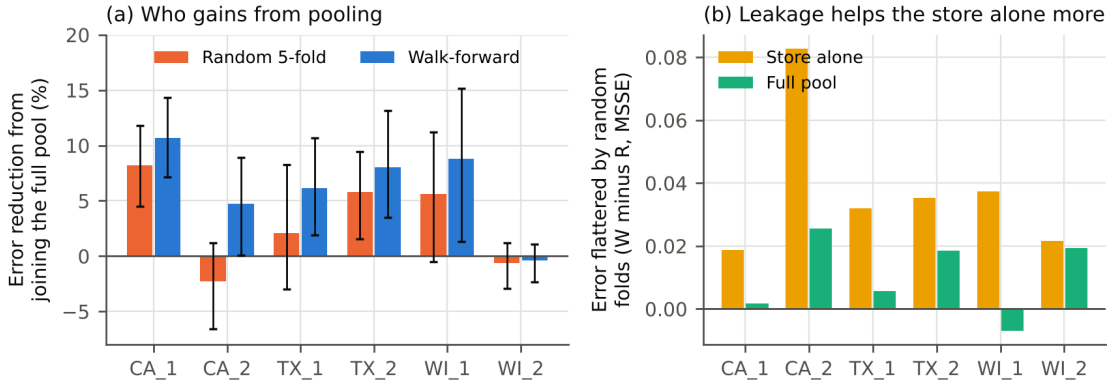


Figure 1. LightGBM pool. (a) Error reduction each store gets from joining the full pool, with bootstrap 95% intervals. (b) How much random folds lower measured error relative to walk-forward, for the store's standalone model and for the full pool.

This is the main result. In the LightGBM game the pool's total value under random folds is less than half its walk-forward value (0.147 against 0.310), and the average store's gain falls from 6.3% to 3.1%. The direction is the same in all five games (Table 1). Figure 1b shows why. Random folds let a model train on weeks that come after the weeks it is tested on, which flatters every model, but not equally: a store's standalone model, trained on little data, benefits on average 0.038 MSSE from the leak, while the pooled model, which already has plenty of data, benefits only 0.011. The same asymmetry holds in every game. Leaked future rows act as a substitute for other firms' data, so a protocol that leaks makes pooling look

less necessary than it is. A cooperative that priced membership with random validation would undersell the pool to its own members.

4.3 The protocol changes who counts as a contributor (Q2)

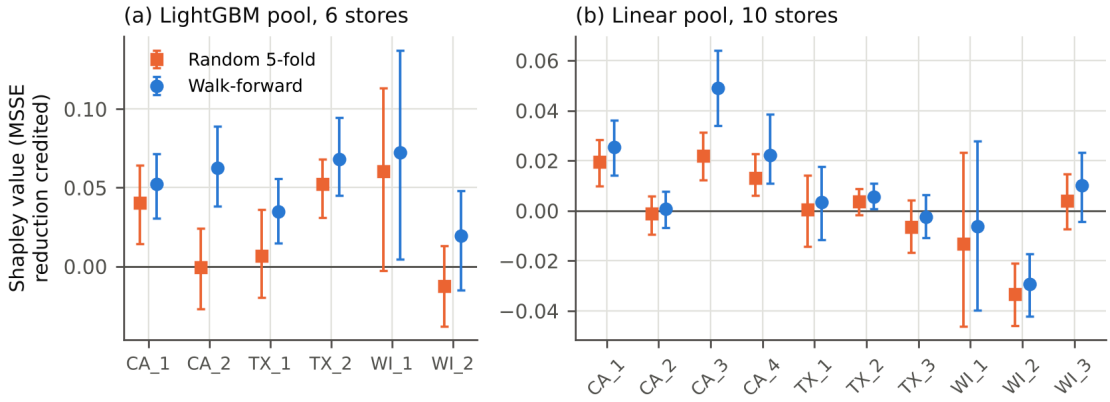


Figure 2. Shapley values with 95% bootstrap intervals under both protocols. (a) LightGBM, six stores. (b) Linear model, ten stores, item sample 1.

Rankings of the stores mostly agree between protocols at the extremes, with Kendall τ between 0.60 and 1.00, but the bootstrap intervals reach as low as 0.07, so the agreement itself is uncertain. The changes that matter for payment are larger than the rank correlations suggest. Under random folds CA_2 looks like a free rider, with a Shapley value of essentially zero and fifth place of six; under walk-forward validation it earns 0.063, the second-largest share (Figure 2a). WI_2 moves from a negative value, meaning the pool would charge it for joining, to a small positive one. Some store's value changes sign in four of the five games. A payout rule based on random validation would therefore not just scale everyone's credit down; it would tell some members they contribute nothing when they contribute a great deal.

4.4 Shapley allocations are never stable (Q3)

An allocation is in the core if no subgroup of firms could do better by leaving and pooling among themselves. Shapley values are outside the core in all ten game-protocol pairs, and the core itself is empty in nine of them, so no allocation at all would keep every subgroup satisfied. In four of the eight linear game-protocol pairs the blocking coalition is WI_2 alone: its Shapley value is negative in all three item samples under both protocols, so Shapley asks it to pay for membership when it could simply forecast alone. The only non-empty core is the LightGBM game under walk-forward validation (least-core $\varepsilon = -0.008$). Random validation would have reported the opposite, an empty core with $\varepsilon = 0.029$. Even there, Shapley lies just outside the core: the five stores other than TX_1 would gain 0.012 (3.7% of the pool's value) by forming their own pool against Shapley payments. In practice this means stability has to be checked, and if necessary enforced with a least-core or nucleolus allocation, rather than assumed from Shapley's fairness axioms.

4.5 Credit drifts, and cheap estimators are good enough (Q4)

Values computed on a single quarter are unstable. WI_1 has the largest credit in six of nine quarters and the smallest in the other three (Figure 3). Across games, the average Kendall τ between consecutive quarters ranges from -0.04 to 0.31. Thirteen weeks of data are too few to separate genuine drift from noise, but either way the practical lesson is the same: credit for a payout should be computed over long rolling windows and reported with intervals, never from one quarter.

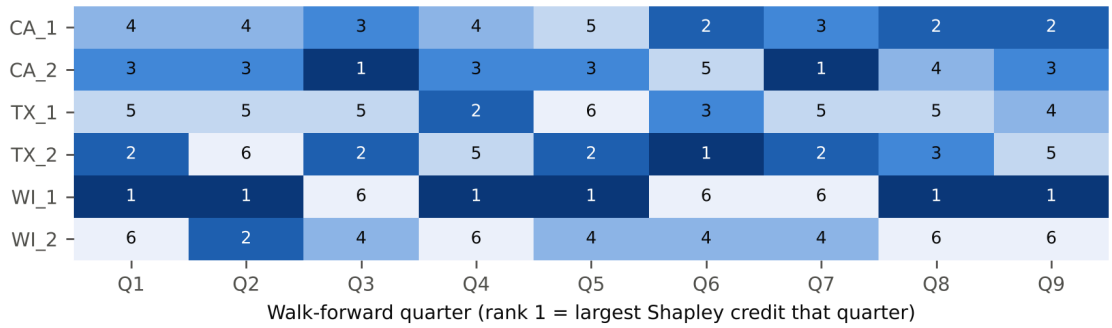


Figure 3. Each store's Shapley rank in each walk-forward quarter of the LightGBM game (1 = largest credit).

Estimator (walk-forward)	Kendall τ with exact Shapley	Relative L1 error	Coalitions evaluated (10 firms)
Leave-one-out	0.78 to 0.96	25% to 66%	11
Monte Carlo, 10 permutations	0.77 to 0.93	15% to 34%	up to 100
Monte Carlo, 50 permutations	0.89 to 0.99	7% to 15%	up to 500
Monte Carlo, 200 permutations	0.95 to 1.00	3% to 9%	up to 1,023

Table 2. Cheap estimators against exact Shapley values, ranges over the five games (Monte Carlo results averaged over 50 repetitions).

For larger pools, where exact computation is impossible, Monte Carlo Shapley with about 200 permutations reproduces the exact ranking ($\tau \geq 0.95$) and magnitudes within 9%. Leave-one-out values rank stores reasonably but misstate how much credit each deserves by a quarter to two thirds, which is too coarse for payments.

5 What this means for a forecasting commons

1. **Value the pool the way it will run.** Walk-forward validation is not a technicality here: random validation halved the measured value of the pool and mislabelled contributors.
2. **Report uncertainty with every payout.** Several stores' intervals cross zero; a member should know whether its credit is distinguishable from nothing.
3. **Check stability, do not assume it.** Shapley was never in the core in these games. A least-core allocation, or an explicit side payment, is needed before firms are asked to commit.
4. **Recompute on long rolling windows** and use Monte Carlo Shapley once the pool is too large for exact computation.

6 Limitations

The ten stores belong to one company and share prices, promotions and suppliers, so they are a proxy for independent firms, not a substitute. Firm sizes were assigned at random to create heterogeneity. The LightGBM game covers six stores, one product sample and one seed; the linear games cover ten stores and three samples. The value function counts only members' own error reductions and treats all stores' errors equally, regardless of sales volume. Forecasts are one week ahead. Quarterly values rest on 13 weeks and are noisy.

7 Next steps

The natural next step is my final-year project's real setting: order histories from independent firms with genuinely different products, where pooling should matter more and stability questions become concrete. Three methodological directions follow from these results: allocation rules that are stable by construction and robust to the validation protocol; strategic behaviour, since a member who knows the valuation rule

may duplicate or withhold data to inflate its credit; and a value function weighted by each firm's economic stake rather than its forecast error alone.

Reproducibility. All code, coalition-level results and the Colab notebook are at `github.com/imadkhan4478/forecast-pool-shapley`. The full study runs in about 15 to 20 minutes on a free Colab CPU.

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